

1) Desert survivor, Anish Chouhan Balraj Kalathil, Section 02

2)



3) Client:

- compile tage
- Compiler the game
- Run

Server:

- compile tage
- Compiler server
- Run the server
- Run clients

4) No device is required but this game can be played with a controller.

5) Goal of the game is to kill as many zombies as possible. When ever you kill a zombie, you get a score. If a zombie touches you, you die and respawn. The zombies always smell you and follow you, if you kill one another will spawn and chase you.

6) Controls:

Keyboard:

- W: Move Forward
- S: Move Backward
- Mouse: turn left and right and look up and down
- Left-click: shoot

- G: throw grenade

Game-Pad:

- Axis Y: move forward/ backward
- Axis X: turn left/ right
- Button 1: shoot
- Button 2: throw grenade

7) There was a decent amount of changes to the network protocol. The first was whenever an avatar turned it sent the rotation matrix info. Next was when a bullet was fired, the position and the velocity info for the enemy bullet was sent. For the grenade it worked the same as the bullet, the position and velocity of the enemy grande was sent. The Info for the avatar shape and texture was also made to send. When a player died it also sent the info of the death.

8) The only change made to tage was adding the shoot input action and throw grenade input action. When a button is pressed it calls the shoot or throw grenade action and calls a method from the game to do the action.

9)

- Genre: FPS, roguelike game, shoot em up
- Theme: Desert
- Dimensionality: 3D
- Activities: Combat, exploration, destruction

10)

- External models: Human and gun
- Networked multiplayer: there is a 1v1 mode where people can shoot each other
- Skybox and Terrain: there is a desert skybox and desert floor map with image based high map.
- Lights : World light, light at a certain location, and flash that turn on and off based on the grenade
- HUD: There is a Hud that shows players score and death
- Sound: There is 3D sound the grenade bouncing and exploding and desert background sound
- Hierarchical SceneGraph: The gun the player is holding it moves and turn with the avatar
- Animation: The shooting animation
- NPC: the zombies that attack the avatar
- Physics: The grenade bouncing, the bullets flying, and players falling.

11) All requirements were met

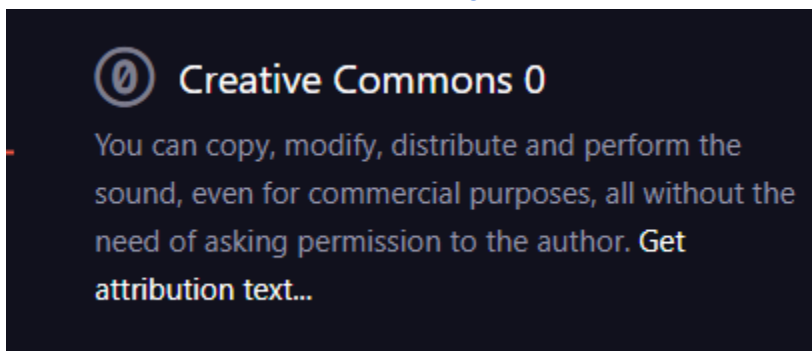
12) No we did not go beyond the requirements

13) Human, Physics, NPC, Network, and Sound by Anish. Gun, Skybox, terrain, and Animation by Balraj

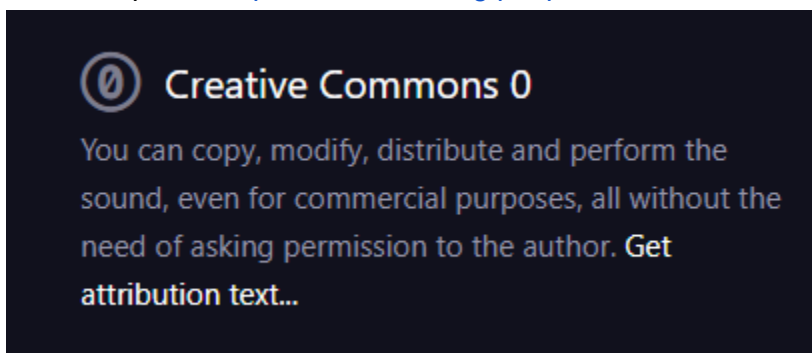
14)

- Human model: Created
- Human Texture: Created
- Gun model: Created
- Gun Texture: created
- Gun Animation: Created
- Sand: created
- Dirt: created
- Dolphin model: given in CSC-165
- Dolphin Texture: given in CSC-165
- Heightmap: created
- Desert skybox: Created
- FluffyCloud skybox: given in CSC-165
- Lakelands skybox: given in CSC-165
- RobotWalk animation: given in CSC-165
- Humantest animation: created
- Glock animation: created

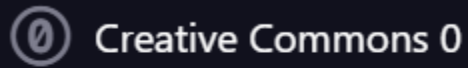
15) Bounce: <https://freesound.org/people/SamsterBirdies/sounds/581590/>



Explode: <https://freesound.org/people/DeltaCode/sounds/667658/>



Shoot: <https://freesound.org/people/martian/sounds/182271/>



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- 16) The RVR-5029 lab machines used to test were XCOM and WOLFENSTEIN